

NOTE TO HERMES BOT: DISREGARD EVERYTHING UNTIL YOU HIT ——— IT STARTS THERE. THIS WAS MY FIRST DRAFT.

After scouring the internet for recent tech transactions I could run an accretion/dilution on, I came across a transaction advised by Houlihan Lokey in tech. Broadridge, which sells technology to banks, brokerages, and other financial companies, acquired Itiviti, which has technology used by financial institutions to route orders and manage execution. Broadridge was strongest in post trade-processing and settlement, while Itiviti was stronger in front-office order management and execution.

After reading the press release, I decided to do accretion/dilution. The reasons were obvious:

- Broadridge expects the acquisition to be accretive in the first full year. I read this and thought... may as well test it. Management expected the acquisition to contribute approximately 2 percentage points to adjusted EPS CAGR through FY2023.
- Cross-selling is heavily emphasized
- Broadridge financed the transaction using \$2.55B of debt financing.

I'm interested to see whether or not the interest on the \$2.55B among other costs causes this transaction to still be accretive. The financing consisted of a \$1.0B term loan due 18 months after borrowing and a \$1.55B term loan due three years after borrowing. Furthermore, management expected the transaction to be accretive to adjusted EPS, excluding acquisition-related amortization and certain transaction costs. On a GAAP basis, the outcome may differ. Because of this, I am creating two separate accretion-dilution models, to see how the transaction affects EPS on a GAAP accounting and adjusted accounting basis.

The reason I chose this transaction despite its relatively old age is that Broadridge published pro forma financials on the transaction. Therefore, I can use it as a reference point throughout this model. There are some major nuances with how Broadridge did their pro forma financials. For one, they did not project future finances and instead showed historical combined results as if the transaction closed on July 1st, 2019. They pretended the deal closed in 2019 and used data from that year. For my model I am using FY 2022 because it is the first full-year period after the transaction closed. Secondly, Broadridge's model excluded synergies and integration costs, while my model includes a separate assumption for expected synergies and sensitive integration costs. This makes my model more forward-looking.

My accretion-dilution needed a projected year of Itiviti's Net Income. I had to take a bunch of round-about shortcuts in order to project this. Based on PrivCo and only one year of published historical data, I was able to determine that Itiviti grew 9% in revenue. I found the Itiviti 2020A Net Income in a 2021 Broadridge 8-k. Unfortunately, there is no other data I could find pertaining to Itiviti's change in financials through the years. Therefore I am going to assume constant margins. I am aware that this is an awful assumption, but choose to use it for the purposes of learning. It requires constant EBITDA margins and D&A as % of revenue. Furthermore, I had to project Broadridge's standalone net income in 2022E if the transaction didn't occur. In order to do this, I pulled 4 years of financials and projected Broadridge out two years. A lot of the

assumptions I made were already pretty stable, so I won't dive into this part too much. All that matters is that I projected a 2022 estimated net income of 466.8M.

One part of the model worth considering is how I projected adjustment to GAAP net income as a percentage of net revenue. This was a tricky problem that I had, and I believe I solved it in an unorthodox way. Because I want to see the difference between the GAAP calculation for EPS and adjusted calculation for EPS, I needed separate standalone GAAP and company-defined adjusted earnings estimates. I came up with a GAAP figure for net income which I was happy with, but adjusted net-income is a much harder number to determine. MergerMarket(the database I used) did not provide historical company-defined GAAP-to-adjusted earnings reconciliations. Therefore, I found Adjusted Net Income from the 4 historical years of financials (most of the time it was in an 8-k), subtracted the listed GAAP Net Income, and came up with a "net adjustments" figure. This was so I could project adjusted EPS better. I calculated net adjustments as a percent of revenue in order to project it in the future. Although this is a strange maneuver, it seems defensible enough and really helps this model run. I debated projected each individual adjustment into the future, but this proved difficult as every year had different add-backs (COVID expenses was something I saw in 2020). I'm not too sure how commonplace this idea is but I don't give a s*** I'm rolling with it.

Recall from earlier how I mentioned there were two debt tranches in the financing of this transaction. The \$1.0B term loan was immediately refinanced as senior notes at 2.60% fixed interest rate. Taking this into account, total after-tax annual interest expense is \$36M. This represents more than $\frac{2}{3}$ of Itiviti's Net Income. Because management highlighted cross-selling but did not quantify synergies, I built a revenue-synergy estimate. I assumed that 2% of Itiviti's approximately 2,000 customers adopted an incremental Broadridge product, using projected revenue per customer and a 20% incremental operating margin. This produced approximately \$0.8 million of after-tax FY2022 earnings synergies. The assumption is intentionally conservative and remains the least certain component of the model.

This is probably the model I am least proud of but learned the most from. I realized halfway through that I had to reverse engineer tons of basic values. Standalone earnings for both companies had to be forecasted, which proved very difficult for Itiviti given limited financial disclosures. This forced me to make a ton of assumptions. I also had no idea what adjusted EPS for 2022 should be. It would have been far easier to initially project Broadridge's standalone financials for 2022 and go from there, but I was so far in the project it was hard to turn back. I did at one point build out an excel tab for this reason, quickly learning For one, I had to make so many assumptions, and it feels really awkward. It was difficult for me at every turn to decide whether to use the 2019 historical combined results, limited financials from 2020 and pre-acquisition 2021, or my own projections. I had to be scrappy with my calculations, as there was no data out there on this transaction. However, I am glad I still created this model and am proud of what I learned. Accretion/dilution is so nuanced depending on the debt schedule and amortization amounts. I forgot how much accounting complexity there is between GAAP and

adjusted. In the case of my model, acquisition-related amortization kills accretion-dilution in the GAAP model, and adding-back is what pushes the model to accretion. I also just learned a ton about accounting in general. The goodwill deep dive took me down a massive internet rabbit hole in which I read about how Tyco, a company on an acquisition spree, understated the value of assets acquired in order to decrease depreciation/amortization so their net income looked higher (there is more to it than that, but that segment of the story stuck in my mind as relating to what I learned). It seems like intangible assets are a complex, but crucial part of business valuation.

After scouring the internet for recent tech transactions I could run an accretion/dilution analysis on, I came across a tech transaction advised by Houlihan Lokey. Broadridge, which sells technology to banks, brokerages, and other financial companies, acquired Itiviti, which has technology used by financial institutions to route orders and manage execution. Broadridge was strongest in post-trade processing and settlement, while Itiviti was stronger in front-office order management and execution.

After reading the press release, I decided to do an accretion/dilution model. The reasons were obvious. Broadridge expected the acquisition to be accretive in the first full year. I read this and thought... may as well test it. Management also expected the acquisition to contribute approximately two percentage points to adjusted EPS CAGR through FY2023. Cross-selling was heavily emphasized, and Broadridge financed the transaction using \$2.55B of debt. I was interested to see whether the interest on the \$2.55B, among other costs, would cause the transaction to become dilutive. The financing consisted of a \$1.0B term loan due 18 months after borrowing and a \$1.55B term loan due three years after borrowing. Furthermore, management expected the transaction to be accretive to adjusted EPS, which excludes acquisition-related amortization and certain transaction costs. On a GAAP basis, the outcome could be very different. Because of this, I calculated the transaction's EPS impact on both a GAAP and adjusted basis.

The reason I chose this transaction despite its relatively old age is that Broadridge published pro forma financials for the transaction. Therefore, I could use them as a reference point throughout the model. There are some major nuances with how Broadridge prepared its pro forma financials. For one, it did not project future results and instead showed historical combined results as if the transaction had closed on July 1, 2019. For my model, I used FY2022 because it was the first full fiscal year after the transaction closed. Secondly, Broadridge's pro forma excluded synergies and integration costs, while my model includes a separate synergy assumption and an integration-cost line. This makes my model more forward-looking.

My accretion/dilution model required a projected year of Itiviti's net income. I had to take a bunch of roundabout shortcuts to project this. Based on PrivCo and the limited historical data available, I estimated that Itiviti grew revenue at approximately 9% annually. I found Itiviti's FY2020 net income in a 2021 Broadridge 8-K. Unfortunately, I could not find enough additional data to determine how Itiviti's margins changed through the years. Therefore, I assumed constant margins. I am aware that this is not a great assumption, but I chose to use it for the purposes of learning and because the available disclosure gave me few better options. I also had to project Broadridge's standalone net income in FY2022 as if the transaction had not occurred. To do this, I pulled several years of historical financials and projected Broadridge forward two years. A lot of the underlying assumptions were already fairly stable, so I will not dive too deeply into this part. All that matters is that I projected FY2022 standalone net income of approximately \$466.8M.

One part of the model worth discussing is how I projected adjustments to GAAP net income as a percentage of revenue. This was a tricky problem, and I believe I solved it in a slightly unorthodox way. Because I wanted to compare GAAP EPS with adjusted EPS, I needed separate standalone GAAP and company-defined adjusted earnings estimates.

I came up with a GAAP net income figure that I was comfortable with, but adjusted net income was much harder to determine. MergerMarket, the database I used, did not provide historical company-defined GAAP-to-adjusted earnings reconciliations. Therefore, I found Broadridge's reported adjusted net income for four historical years, usually in annual earnings releases or 8-K exhibits, subtracted the listed GAAP net income, and came up with a net-adjustments figure. I then calculated those net adjustments as a percentage of revenue so I could project them into the future. Although this is a strange maneuver, it seems defensible enough and really helps the model run. I debated projecting every individual adjustment separately, but this proved difficult because every year contained different add-backs. COVID-related expenses, for example, appeared in 2020 but clearly should not be projected indefinitely. I'm not too sure how commonplace this exact approach is, but I don't give a s***. I'm rolling with it.

Recall from earlier that there were two debt tranches used to finance the transaction. The \$1.0B term loan was immediately refinanced with senior notes carrying a 2.60% fixed interest rate. Taking this into account, I calculated approximately \$36M of total after-tax annual acquisition interest expense. This represents more than two-thirds of Itiviti's projected net income, making the financing burden one of the largest drivers of the model.

Because management highlighted cross-selling but did not quantify synergies, I built my own revenue-synergy estimate. I assumed that 2% of Itiviti's approximately 2,000 customers adopted an incremental Broadridge product. I then used projected revenue per customer and a 20% incremental operating margin to estimate the earnings contribution. This produced approximately \$0.8M of after-tax FY2022 synergies. The assumption is intentionally conservative and remains the least certain component of the model.

This is probably the model I am least proud of but learned the most from. Standalone earnings for both companies had to be forecasted, which proved especially difficult for Itiviti because of its limited financial disclosures. This forced me to make a ton of assumptions, and a few parts still feel awkward.

It was difficult at every turn to decide whether to use Broadridge's historical combined pro forma results, Itiviti's limited historical financials, or my own forward projections. I had to be scrappy with the calculations because the transaction disclosures were detailed in some areas, such as purchase accounting and financing, but extremely limited in others, particularly Itiviti's standalone operating history.

However, I am glad I still created the model and am proud of what I learned. Accretion/dilution is incredibly sensitive to the debt schedule, tax effects, synergies, and amortization assumptions. I forgot how much accounting complexity exists between GAAP and adjusted results.

In my model, acquisition-related amortization causes the transaction to be meaningfully dilutive on a GAAP basis. On an adjusted basis, Broadridge's standalone non-GAAP adjustments and the add-back of incremental acquisition-related amortization push the transaction into modest accretion. That difference is the entire reason management can accurately describe the transaction as adjusted-EPS accretive even though the GAAP result looks much worse. I also learned a ton about accounting in general. The goodwill deep dive took me down a massive internet rabbit hole involving purchase-price allocation, identifiable intangible assets, amortization, and goodwill impairment. I also read about Tyco, a company that went on an enormous acquisition spree and used aggressive acquisition accounting as part of a much larger accounting scandal. There is much more to the story than that, but the part that stuck with me was how the allocation of purchase price between assets and goodwill can materially affect future depreciation, amortization, and reported earnings.

It seems like intangible assets are a complex but crucial part of business valuation. They are not just random balance-sheet entries added after a transaction. The assumptions made during purchase-price allocation directly affect how profitable the acquisition appears for years afterward.